

1(a)(i)	work done per unit mass	B1
	idea of work done moving mass from infinity (to the point)	B1
1(a)(ii)	(gravitational) force is attractive	B1
	(gravitational) potential at infinity is zero	B1
	decrease in potential energy as masses approach or displacement and force in opposite directions	B1
1(b)(i)	<i>Either</i> $mv^2 / R = GMm / R^2$ <i>Or</i> $v = \sqrt{GM / R}$ $v^2 = (6.67 \times 10^{-11} \times 6.00 \times 10^{24}) / (7.30 \times 10^6)$	C1
	giving $v = 7.4 \times 10^3 \text{ m s}^{-1}$	A1
1(b)(ii)	$V_P = -GMm / R$	C1
	$= - (6.67 \times 10^{-11} \times 6.00 \times 10^{24} \times 340) / (7.30 \times 10^6)$	C1
	$V_P = -1.9 \times 10^{10} \text{ J}$	A1
1(c)	$v^2 \propto 1 / r$, (r smaller) so v greater	M1
	and E_K greater	A1